SOC-Based Threat Detection & Incident Response Lab

A practical implementation of a Security Operations Center (SOC) lab for threat detection and automated incident response.

Project Objective:

- To design and implement a centralized Security Operations Center (SOC) lab.
- To simulate real-world cyber attacks in a controlled virtual environment.
- To detect attacks using SIEM and IDS tools.
- To implement automated incident response mechanisms.

Technologies Used:

- Wazuh (SIEM)
- Suricata (Network IDS)
- Zeek (Network Analysis)
- Apache2 (Web Server)
- ModSecurity (Web Application Firewall)
- Kali Linux (Attack Simulation)
- VirtualBox (Virtual Environment)

Project Phases:

- Phase 1 – Infrastructure Setup
- Phase 2 – Detection Layer Configuration
- Phase 3 – Attack Simulation & Log Analysis
- Phase 4 – Automated Incident Response

Implemented by

ETTERI GIRI BABU

Email: girishkumar346272@gmail.com

PHASE 1 – INFRA SETUP

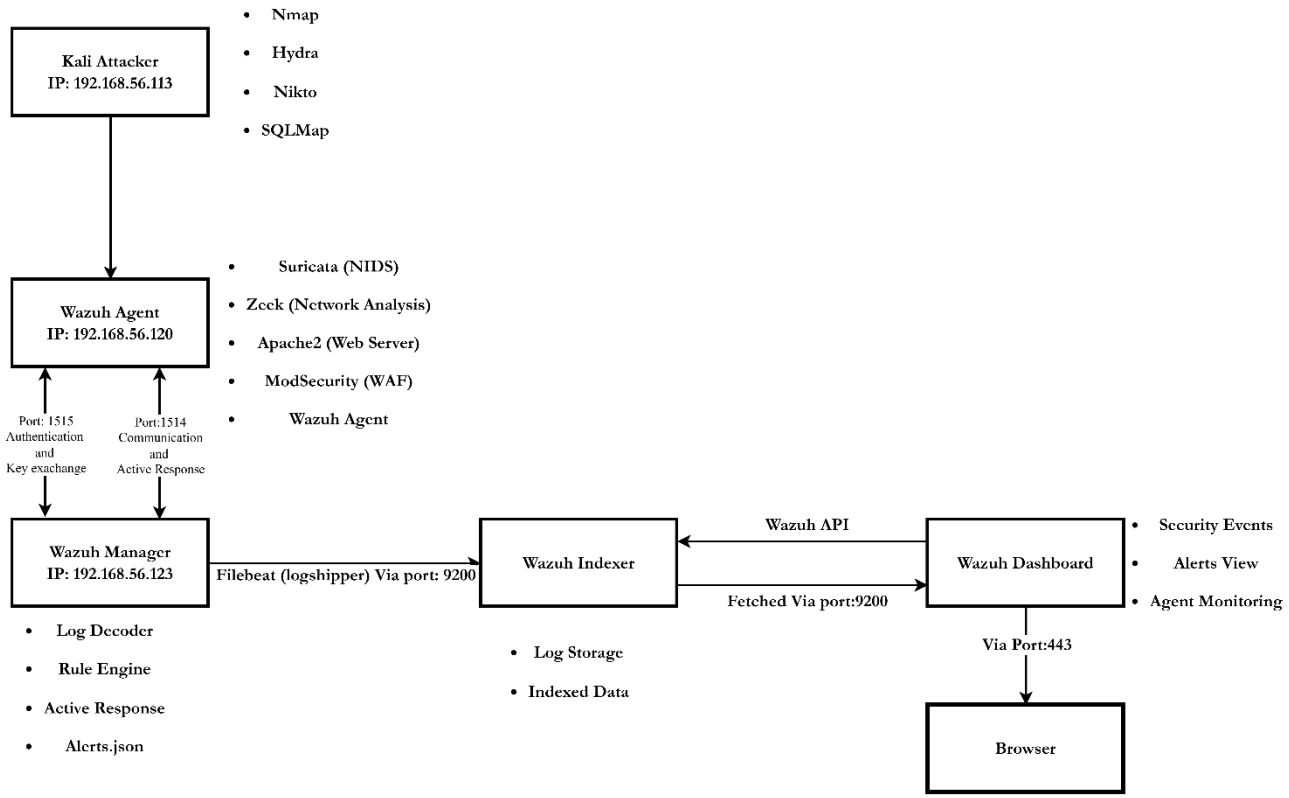
Virtual Lab Setup

Machine	Role	IP Address
Kali Attacker (Kali)	Attacker	192.168.56.113
Wazuh Manager (Ubuntu)	SIEM Server	192.168.56.123
Wazuh Agent (Ubuntu)	Victim Machine	192.168.56.120

Network Configuration:

- All VMs were connected using **Host-Only Network** in VirtualBox.
- All systems were configured in the same subnet (192.168.56.0/24).
- Connectivity verified using ping command.

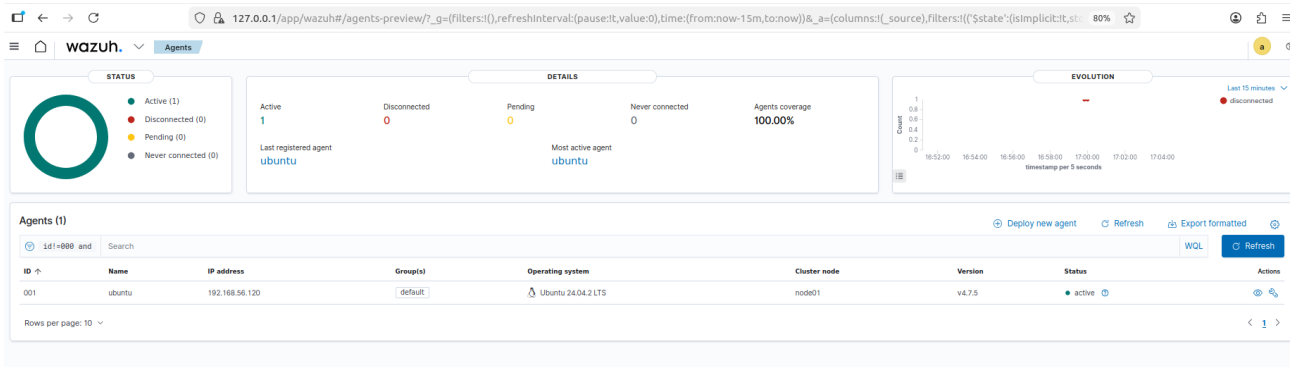
Architecture of the project



Wazuh Manager Installation

- Installed Wazuh on Manager VM.
- All-in-one installation was used.
- Verified services:
  - wazuh-manager
  - wazuh-indexer
  - wazuh-dashboard
- Dashboard accessed via HTTPS (Port 443).

```
oxuser@Ubuntu:~/Desktop$ sudo systemctl list-units --type=service | grep wazuh
wazuh-dashboard.service                          loaded active running wazuh-dashboard
wazuh-indexer.service                            loaded active running Wazuh-indexer
wazuh-manager.service                            loaded active running Wazuh manager
oxuser@Ubuntu:~/Desktop$
```



Wazuh Agent Installation

- Installed Wazuh Agent on Victim VM.
- Registered agent with Manager using port 1515.
- Encrypted communication established via port 1514.
- Verified connection using agent_control -l.

```
vboxuser@Ubuntu:~/Desktop$ sudo tail -f /var/ossec/logs/ossec.log
2026/03/01 11:36:54 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/apache2/error.log'.
2026/03/01 11:36:54 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/auth.log'.
2026/03/01 11:36:54 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/suricata/eve.json'.
2026/03/01 11:36:54 wazuh-logcollector: ERROR: (1103): Could not open file '/var/log/snort/alert' due to [(
2026/03/01 11:36:54 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/snort/alert'.
2026/03/01 11:36:54 wazuh-logcollector: INFO: (1950): Analyzing file: '/var/log/apache2/modsec_audit.log'.
2026/03/01 11:36:54 wazuh-logcollector: INFO: Started (pid: 1810).
2026/03/01 11:37:04 wazuh-syscheckd: INFO: (6009): File integrity monitoring scan ended.
2026/03/01 11:37:04 wazuh-syscheckd: INFO: FIM sync module started.
2026/03/01 11:38:08 rootcheck: INFO: Ending rootcheck scan.
```

PHASE 2 DETECTION LAYER CONFIGURATION

Suricata Installation

- Installed Suricata on Agent VM.
- Configured correct network interface.
- Enabled rule updates using suricata-update.
- Configured Wazuh Agent to forward Suricata logs to Manager.

```
vboxuser@Ubuntu:~/Desktop$ tail -1 /var/log/suricata/eve.json
{"timestamp":"2026-03-01T11:48:18.250264+0000","event_type":"stats","stats":{"uptime":690,"capture":{"kernel_packets":3701,"kernel_drops":0,"errors":0,"afp
acket":{"busy_loop_avg":0,"polls":38625,"poll_signal":0,"poll_timeout":35686,"poll_data":2939,"poll_errors":0,"send_errors":0}},"decoder":{"pkts":3701,"byt
es":1064499,"invalid":0,"ipv4":3615,"ipv6":67,"ethernet":3701,"arp":19,"unknown_ethertype":0,"chdlc":0,"raw":0,"null":0,"sll":0,"tcp":3574,"udp":77,"sctp":
0,"esp":0,"icmpv4":0,"icmpv6":23,"ppp":0,"pppoe":0,"geneve":0,"gre":0,"vlan":0,"vlan_qinq":0,"vlan_qinqinq":0,"vxlan":0,"vntag":0,"ieee8021ah":0,"teredo":0
,"ipv4_in_ipv6":0,"ipv6_in_ipv6":0,"mpls":0,"avg_pkt_size":287,"max_pkt_size":1514,"max_mac_addrs_src":0,"max_mac_addrs_dst":0,"erspan":0,"nsh":0,"event":{
"ipv4":{"pkt_too_small":0,"hlen_too_small":0,"iplen_smaller_than_hlen":0,"trunc_pkt":0,"opt_invalid":0,"opt_invalid_len":0,"opt_malformed":0,"opt_pad_requi
red":8,"opt_eol_required":0,"opt_duplicate":0,"opt_unknown":0,"wrong_ip_version":0,"icmpv6":0,"frag_pkt_too_large":0,"frag_overlap":0,"frag_ignored":0},"ic
mpv4":{"pkt_too_small":0,"unknown_type":0,"unknown_code":0,"ipv4_trunc_pkt":0,"ipv4_unknown_ver":0},"icmpv6":{"unknown_type":0,"unknown_code":0,"pkt_too_sm
all":0,"ipv6_unknown_version":0,"ipv6_trunc_pkt":0,"mld_message_with_invalid_hl":0,"unassigned_type":0,"experimentation_type":0},"ipv6":{"pkt_too_small":0,
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_dupl_eh":0,"exthdr_invalid_optlen":0,"wrong_ip_version":0,"exthdr_ah_res_not_null":0,"hopopts_unknown_opt":0,"hopopts_only_padding":0,"dstopts_unknown_opt
":0,"dstopts_only_padding":0,"rh_type_0":0,"zero_len_padn":15,"fh_non_zero_reserved_field":0,"data_after_none_header":0,"unknown_next_header":0,"icmpv4":0,
"frag_pkt_too_large":0,"frag_overlap":0,"frag_invalid_length":0,"frag_ignored":0,"ipv4_in_ipv6_too_small":0,"ipv4_in_ipv6_wrong_version":0,"ipv6_in_ipv6_to
o_small":0,"ipv6_in_ipv6_wrong_version":0},"tcp":{"pkt_too_small":0,"hlen_too_small":0,"invalid_optlen":0,"opt_invalid_len":0,"opt_duplicate":0},"udp":{"pk
t_too_small":0,"hlen_too_small":0,"hlen_invalid":0,"len_invalid":0},"sll":{"pkt_too_small":0},"ethernet":{"pkt_too_small":0},"ppp":{"pkt_too_small":0,"vju_
pkt_too_small":0,"ip4_pkt_too_small":0,"ip6_pkt_too_small":0,"wrong_type":0,"unsup_proto":0},"pppoe":{"pkt_too_small":0,"wrong_code":0,"malformed_tags":0},
```
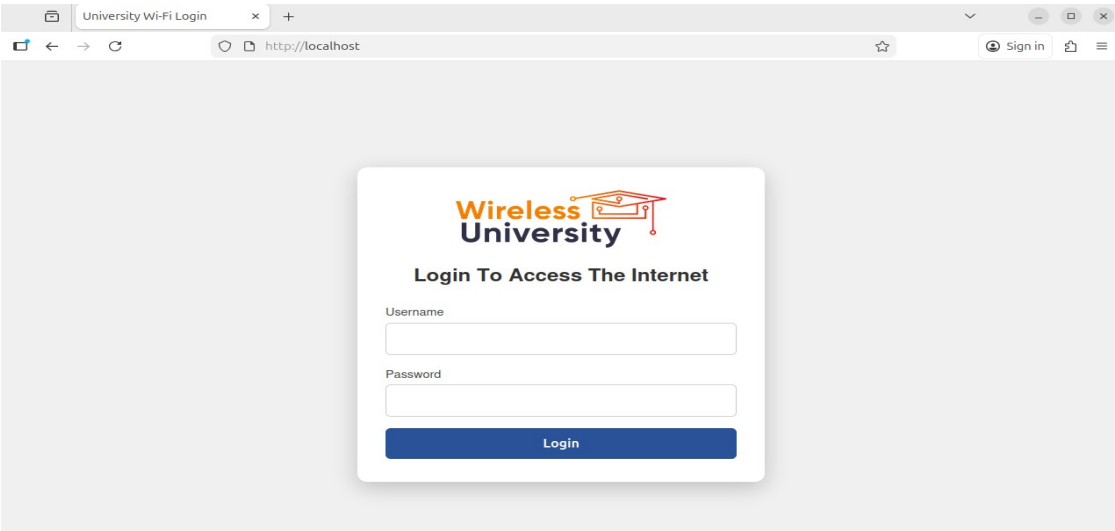

Zeek Configuration

- Installed Zeek on Agent VM.
- Configured standalone mode with correct interface.
- Generated logs such as
- Used Zeek logs for behavioral traffic analysis.

Apache Installation

- Installed Apache2 on Agent VM.
- Hosted test web application (login.php).
- Verified access from Kali machine.



ModSecurity (Web Application Firewall)

- Installed ModSecurity module for Apache.
- Enabled rule engine.
- Forwarded ModSecurity logs to Wazuh Manager.

```
tail: cannot open '/var/log/apache2/modsec_audit.log' for reading: Permission denied
vboxuser@Ubuntu:~/Desktop$ sudo tail /var/log/apache2/modsec_audit.log
Apache-Handler: application/x-httpd-php
Stopwatch: 1772197118000037 11481 (- - -)
Stopwatch2: 1772197118000037 11481; combined=6529, p1=854, p2=4119, p3=79, p4=697, p5=780, sr=101, sw=0, l=0, gc=0
Response-Body-Transformed: Dechunked
Producer: ModSecurity for Apache/2.9.7 (http://www.modsecurity.org/); OWASP_CRS/3.3.5.
Server: Apache/2.4.58 (Ubuntu)
Engine-Mode: "ENABLED"

--fbc3791d-Z--

vboxuser@Ubuntu:~/Desktop$
```

Log Forwarding to Wazuh Manager

Configured Wazuh Agent (ossec.conf) to monitor:

- Apache access and error logs
- Suricata eve.json
- ModSecurity audit logs
- Zeek connection logs

PHASE 3 – ATTACK LAYER

Reconnaissance Attack (Port Scanning)

Performed TCP SYN scan and service enumeration against the victim machine.

Detection:

- Suricata generated scan alerts.
- Wazuh detected suspicious network activity.

Analysis:

- Multiple ports were probed in short duration.
- Source IP: 192.168.56.113
- Alert level generated in Wazuh Dashboard.

SSH Brute Force Attack

Performed password brute-force attack on SSH service.

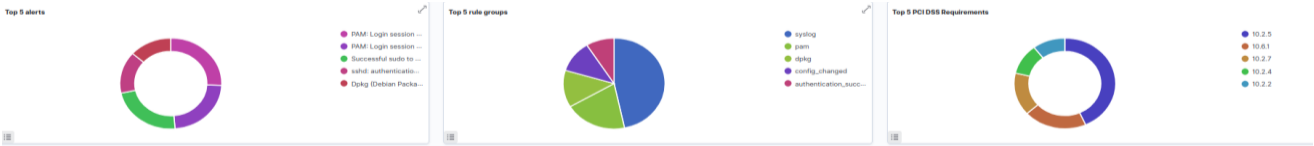
```
┌──(kali㉿kalii)-[~/Desktop]
└─$ sudo hydra -l vboxuser -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.120
sudo: unable to resolve host kalii: Name or service not known
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, o
r for illegal purposes (this is non-binding, these ***** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-02 01:17:40
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort ... (use option -I to skip waiting)) from a previous session found, to
 prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.56.120:22/
[STATUS] 55.00 tries/min, 55 tries in 00:01h, 14344347 to do in 4346:47h, 13 active
^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.
```

Detection:

- Multiple failed login attempts recorded in auth.log. Wazuh rule triggered for auth failure.
- Alert mapped to MITRE ATT&CK – Brute Force (T1110).

Analysis:

- Repeated failed login attempts from attacker IP.
- High alert level generated.
- Security event visible in dashboard.



Security Alerts

Time	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 1, 2026 @ 17:57:02.512	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 17:57:02.507	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 17:57:02.507	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 17:57:02.507	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 17:57:02.503	T1110	Credential Access	sshd: brute force trying to get access to the system. Authentication failed.	10	5763
Mar 1, 2026 @ 17:57:02.446	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 1, 2026 @ 17:57:00.440	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760

Web Application Attack

Simulated web vulnerability scanning and SQL injection attempts on login.php using nikto and sqlmap

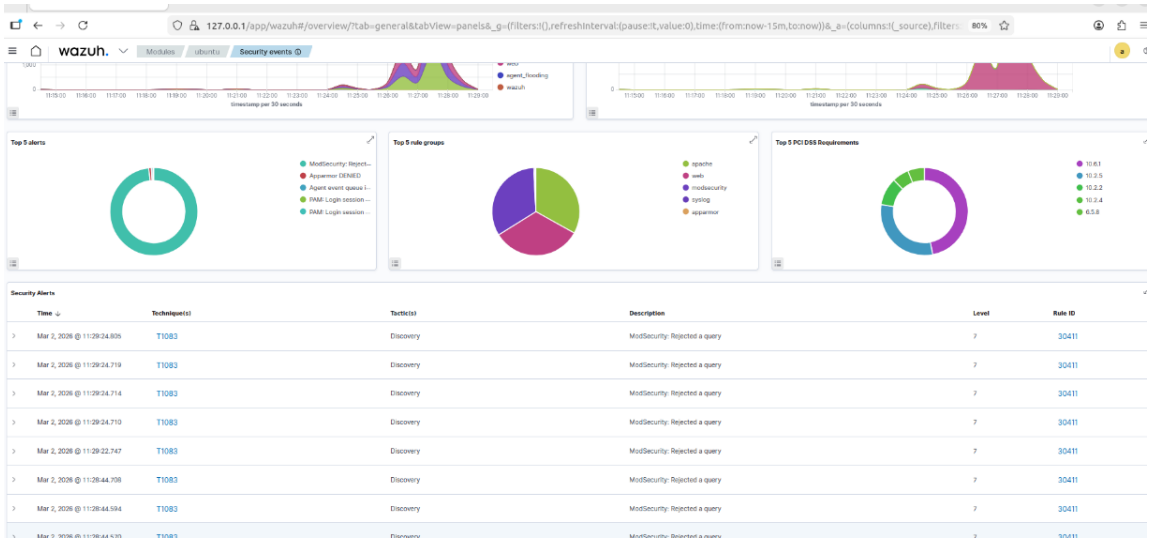
```
┌──(kali㉿kali)-[~/Desktop]
└─$ nikto -h http://192.168.56.120
- Nikto v2.5.0
---------------------------------------------------------------------------
+ Target IP:          192.168.56.120
+ Target Hostname:    192.168.56.120
+ Target Port:        80
+ Start Time:         2026-03-02 00:55:50 (GMT-5)
---------------------------------------------------------------------------
+ Server: Apache/2.4.58 (Ubuntu)
+ /: Cookie PHPSESSID created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/
Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a
 different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-co
ntent-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /backup.tar.lzma: Potentially interesting backup/cert file found. . See: https://cwe.mitre.org/data/definitions/530.htm
l
```

Detection:

- Apache access logs recorded suspicious POST requests.
- ModSecurity generated audit logs for malicious payload.
- Wazuh correlated and displayed alerts.

Analysis:

- Suspicious HTTP requests observed.
- SQL injection patterns detected when ModSecurity enabled.
- Alerts visible in dashboard.



Behavioral Investigation Using Zeek

- Reviewed conn.log for abnormal connection patterns.
- Identified rapid connection attempts during Nmap scan.
- Analyzed HTTP request behavior during web attack.

This demonstrates behavioral analysis in addition to signature-based detection

PHASE 4 – ACTIVE RESPONSE LAYER

Active Response Configuration

Configured Active Response in Wazuh Manager.

```
  <command>
    <name>netsh</name>
    <executable>netsh.exe</executable>
    <timeout_allowed>yes</timeout_allowed>
  </command>

  <active-response>
<command>firewall-drop</command>
<location>all</location>
<level>7</level>
<timeout>600</timeout>
  </active-response>

  <!-- Log analysis -->
```

Configured firewall-drop command to trigger when brute force rules are matched.

- firewall-drop → Blocks attacker IP using iptables.
- location local → Executes on Agent machine.
- rules_id → Triggers for brute-force detection.
- timeout 600 → Blocks attacker for 10 minutes.

Attack After Configuration

Performed SSH brute-force attack again from Kali machine using Hydra.

Observed:

- Wazuh detected repeated authentication failures.
- Active Response executed automatically.
- Attacker IP (192.168.56.113) blocked.

```
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo hydra -l vboxuser -P /usr/share/wordlists/rockyou.txt ssh://192.168.56.120
sudo: unable to resolve host kalii: Name or service not known
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^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.

┌──(kali㉿kali)-[~/Desktop]
└─$ ping 192.168.56.120
PING 192.168.56.120 (192.168.56.120) 56(84) bytes of data.
^C
--- 192.168.56.120 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3414ms


┌──(kali㉿kali)-[~/Desktop]
└─$
```

Verification of IP Blocking

On Agent machine:

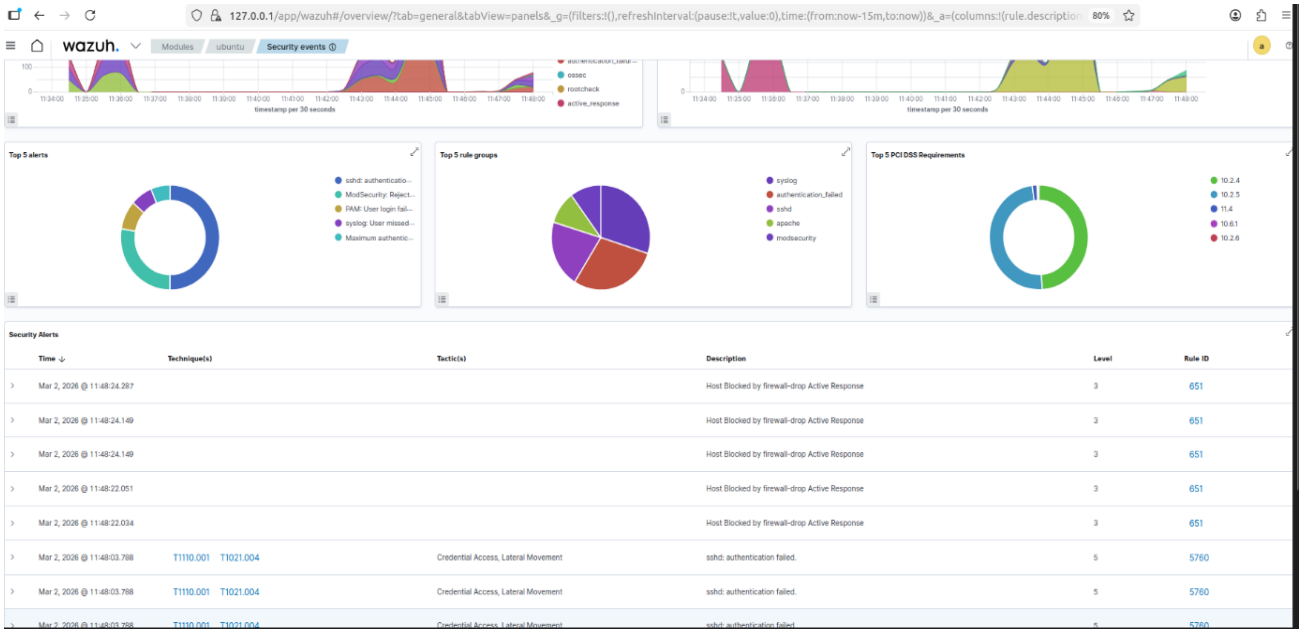
- Connection attempts blocked.

```
vboxuser@Ubuntu:~/Desktop$ sudo iptables -L -n -v
[sudo] password for vboxuser:
Chain INPUT (policy ACCEPT 35901 packets, 5048K bytes)
 pkts bytes target     prot opt in     out     source               destination
  230 23296 DROP       0    --  *      *       192.168.56.113       0.0.0.0/0

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target     prot opt in     out     source               destination
    0     0 DROP       0    --  *      *       192.168.56.113       0.0.0.0/0

Chain OUTPUT (policy ACCEPT 42188 packets, 47M bytes)
 pkts bytes target     prot opt in     out     source               destination

Chain DOCKER (0 references)
 pkts bytes target     prot opt in     out     source               destination
```

Dashboard Alert

Wazuh Dashboard displayed:

- Alert for brute force
- Notification of Active Response execution
- Host blocked event



Conclusion

This project successfully demonstrated the design and implementation of a centralized SOC lab using Wazuh integrated with Suricata and Zeek.

A controlled virtual environment was created to simulate real-world cyber attacks such as reconnaissance, brute-force, and web-based attacks. These attacks were successfully detected using SIEM and IDS tools. Logs from network, host, and application layers were centralized and analyzed through the Wazuh Dashboard.

Finally, automated incident response was implemented using Active Response, which dynamically blocked malicious IP addresses upon detection.

This project provided practical understanding of:

- SIEM architecture and log flow
- Multi-layer threat detection
- Security event correlation
- Automated incident response mechanisms